

Lab 3: Architectural Pattern Selection & Justification

Alumni Mentorship & Mock Interview Platform

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Architecture Selection: "We chose Microservices Architecture for the Alumni Mentorship & Mock Interview Platform System."

Reason 1 — Independent, Demand-Driven Scaling

NFR-002 requires the platform to keep end-to-end API response times under 2 seconds for search, recommendation, and slot-retrieval requests even under a concurrent load of 1,000 users, a load that spikes sharply during campus placement drives. A microservices architecture lets the Mentor Discovery & Matching service (search/filter/recommend) and the Booking & Scheduling service scale out independently on their own instances, while low-traffic services such as Feedback & Scorecards stay at baseline capacity. A layered or client-server design would force the entire application tier (or the single server) to be scaled together, wasting resources and making the 2-second SLA far harder to guarantee at peak load.

Reason 2 — Independent Deployment Aligned to Functional Domains

The system naturally decomposes into five independent functional domains — Mentor Discovery, Booking & Availability, Session Notifications, Interview Feedback, and Profile/Privacy — the same epics already used for backlog and sprint planning in Lab 2. Microservices let each domain be built, tested, and deployed on its own release cycle without redeploying the whole platform, and they isolate failure: if the external Calendar/Notification service times out (the 7a alternate flow in the Lab 1 use-case), only the Notification service is affected, while Booking, Discovery, and Feedback continue operating normally.

Security Advantage

NFR-001 requires all profile data and feedback scorecards to be soft-deleted within 24 hours of a request and to remain compliant with privacy guidelines. Isolating this responsibility inside a single User Profile & Privacy service means personally identifiable data and access-control/soft-deletion logic live behind one well-defined boundary (ProfileInterface) instead of being scattered across the codebase. This limits the blast radius of a compromise: an attacker who breaches the Notification service, for example, gains no direct access to profile or scorecard data, since every other service must go through the Profile service's controlled interface to reach it.

Performance Benefit

Because Mentor Discovery & Matching is a separate component, it can be given its own read-optimized data path (e.g., caching or a read replica of mentor profiles) to serve high-volume search and recommendation traffic quickly, without competing for resources with the write-heavy Booking & Scheduling service that must guarantee no double-booking. This separation directly supports the NFR-002 target of a sub-2-second response time under 1,000 concurrent users, something a single shared-tier layered architecture would struggle to isolate.